

W[5] generates all connected Feynman diagrams

• Contains redundancies such as

$$\text{---} \bigcirc \text{---} \bigcirc \text{---} \sim \text{---} \bigcirc \times \bigcirc \text{---}$$

$$\text{---} \bigcirc \text{---} = \sum \left(\text{all connected diagrams with } n\text{-external legs} \right)$$

$$\text{---} \bigcirc \text{---} \equiv \sum \left(\text{irreducible diagrams with } n \text{ external legs computed} \right)$$

$$\text{---} \bigcirc \text{---} = \text{---} \bigcirc \text{---} \bigcirc \text{---} \bigcirc \text{---}$$

$\text{---} \bigcirc \text{---}$ (irreducible cubic vertex) $V_3(a, b, c)$
 $\text{---} \bigcirc \text{---}$ (correctors to external legs)

* Going from Connected to 1PI is given by the Legendre transformation of W[5]

$$\frac{\partial (iW[5])}{\partial J(x)} = \langle \Omega | \hat{\phi}(x) | \Omega \rangle \equiv \bar{\phi}(x)$$

Define

$$\Gamma(\bar{\phi}) = W[5] - \int d^4x J(x) \bar{\phi}(x)$$

(Similar to $L = -H + p\dot{x}$)

• Idea is $\frac{\delta \Gamma(\bar{\phi})}{\delta J(x)} = \underbrace{\frac{\delta W[5]}{\delta J(x)}}_{\bar{\phi}(x)} - \bar{\phi}(x) = 0$

• Or we solve:

$$\boxed{\Phi(x) = \delta W[\bar{\psi}] / \delta \bar{\psi}(x)}$$

for $J(x)$ in terms of $\Phi(x)$ and substitute back

$$\begin{aligned} \frac{\delta \Gamma(\bar{\psi})}{\delta \bar{\psi}(x)} &= \frac{\delta W[\bar{\psi}]}{\delta \bar{\psi}(x)} - J(x) - \int d^d z \frac{\delta J(z)}{\delta \bar{\psi}(x)} \bar{\psi}(z) \\ &= -J(x) + \int d^d z \left(\frac{\delta W[\bar{\psi}]}{\delta \bar{\psi}(z)} \frac{\delta J(z)}{\delta \bar{\psi}(x)} - \frac{\delta J(z)}{\delta \bar{\psi}(x)} \bar{\psi}(z) \right) \\ &= -J(x) \end{aligned}$$

$$\boxed{\frac{\delta \Gamma(\bar{\psi})}{\delta \bar{\psi}(x)} = -J(x)}$$

* Now, consider the two point fn

$$\begin{aligned} G_2(x, y) &= i(-i)^2 \frac{\delta^2}{\delta J(x) \delta J(y)} W[\bar{\psi}] \\ &= -i \frac{\delta}{\delta J(x)} \left(\bar{\psi}(y) \right) = \boxed{(-i) \frac{\delta \bar{\psi}(y)}{\delta J(x)}} \quad \text{★} \end{aligned}$$

* Functional Taylor Series

$$\boxed{\Gamma(\bar{\psi}) = \sum_{k=0}^{\infty} \frac{1}{k!} \int d^d x_1 \dots \int d^d x_k \left(\prod_{i=1}^k \bar{\psi}(x_i) \right) \Gamma_k(x_1, \dots, x_k)}$$

$$\Gamma_k(x_1, \dots, x_k) = \frac{\delta^k \Gamma(\bar{\psi})}{\delta \bar{\psi}(x_1) \dots \delta \bar{\psi}(x_k)} \Big|_{\bar{\psi}=0}$$

$$\therefore \Gamma_2(x, y) = \frac{\delta^2 \Gamma(\bar{\psi})}{\delta \bar{\psi}(x) \delta \bar{\psi}(y)} = \boxed{-\frac{\delta J(y)}{\delta \bar{\psi}(x)}} \quad \text{★}$$

Thus

$$\int d^d z \Gamma_2(x, z) G_2(z, y) = \int d^d z \left(-\frac{\delta J(x)}{\delta \bar{\psi}(z)} \right) \left(-i \frac{\delta \bar{\psi}(z)}{\delta J(y)} \right)$$

$$\int d^d z \Gamma_2'(x, z | G_2(z, y)) = i \delta^d(x-y)$$

Formally

$$G_2(x, y) = i \Gamma_2'^{-1}(x, y)$$

* To better understand this go to momentum space

$$G_2^f(x, y) = \int \frac{d^d k}{(2\pi)^d} \frac{-i}{k^2 + m^2 - i\epsilon} e^{ik(x-y)}$$

Thus

$$\begin{aligned} \Gamma_2'(x, y) &= - \int \frac{d^d k}{(2\pi)^d} (k^2 + m^2 - i\epsilon + \Sigma(k)) e^{ik(x-y)} \\ &= - \int \frac{d^d k}{(2\pi)^d} e^{ik(x-y)} (k^2 + m^2 - i\epsilon) \\ &\quad - \int \frac{d^d k}{(2\pi)^d} \Sigma(k) e^{ik(x-y)} \end{aligned}$$

$\Rightarrow \Gamma_2'$ describes the self-energy corrections

$$\begin{aligned} \Gamma_2'(k) &= \Gamma_2^{\text{free}} - \Sigma(k) \\ &= \Gamma_2^f (1 - (\Gamma_2^f)^{-1} \Sigma(k)) \end{aligned}$$

$$\begin{aligned} G_2(k) &= i \left(\Gamma_2^f - \Sigma(k) \right)^{-1} = i (1 - (\Gamma_2^f)^{-1} \Sigma(k)) (\Gamma_2^f)^{-1} \\ &= (1 - i G_2^f \Sigma)^{-1} G_2^f \end{aligned}$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$G_2(k) = G_2^f + G_2^f \Sigma G_2^f + i^2 G_2^f \Sigma G_2^f \Sigma G_2^f + \dots$$

which is the summation

$$\begin{aligned} \text{---} \textcircled{||} \text{---} &= \text{---} + \text{---} \textcircled{##} \text{---} \\ &+ \text{---} \textcircled{##} \textcircled{##} \text{---} + \dots \end{aligned}$$

↑ 1PI self energy

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